Unit 2

WHOLE NUMBERS

2.1 NATURAL AND WHOLE NUMBERS

We use numbers '1, 2, 3, 4, 5, ...' for representing the quantity of objects in our daily life.

These numbers 1, 2, 3, 4, ..., are called natural numbers.

When we include '0' in the set of natural numbers, we will get the set of whole numbers.

'0, 1, 2, 3, 4, ...' are called whole numbers.

Here 0 is predecessor of 1 and 1 is successor of 0. Similarly 1 is predecessor of 2 and 2 is successor of 1 and so on.

Differentiate between natural and whole numbers

The only difference between set of natural numbers and set of whole numbers is of number '0', '0' does not belong to the set of Natural numbers. All the other members of both the sets are the same.

Identify natural and whole numbers, and their notations

We denote the set of natural numbers by N.

i.e $N = \{1, 2, 3, 4, 5, 6, ...\}$

The set of whole numbers consist of zero and all natural numbers. It is denoted by W,

i.e. $W = \{0, 1, 2, 3, 4, 5, 6, 7, ...\}$

Look at the following sets:

$N = \{1, 2, 3, ...\}$

$W = \{0, 1, 2, ...\}$

Important facts about natural numbers and whole numbers are:

- 1 is the smallest and the first natural number.
- The first and the smallest whole number is '0'.
- The set of natural numbers is an infinite as we can not count all natural numbers.
- The set of whole numbers is an infinite set.

Represent a given list of whole numbers

Number line helps us to represent given whole numbers. Whole numbers can be represented on a number line by using following steps:

Let us represent numbers 0, 1, 2, 3, 4, 5, ..., on a number line.

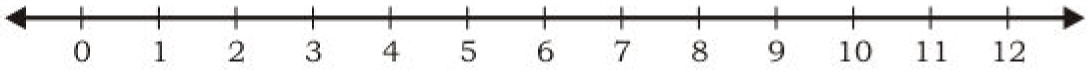
Steps:

(i) Draw a line 

(ii) Mark a point to represent the first whole number '0' on the line.

(iii) Mark other points at equal distance and name them 1, 2, 3, 4, 5, ... respectively as shown below:

Thus we get a number line as under:



Note:

(i) On a number line, a number is greater than any number on its left. For example, $1 > 0$, $2 > 1$ and $3 > 2$ etc.

(ii) On a number line a number is less than any number on its right. For example $0 < 2$, $1 < 3$, $5 < 12$.

Represent whole numbers $<$ (or $>$) a given whole number on a number line

Let us represent whole numbers $<$ (or $>$) a given whole number on number line with the help of an example.

Example.

Represent the following on number line.

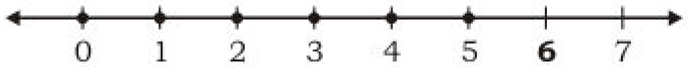
(i) Whole numbers Less than 6

(ii) Whole numbers Greater than 4

Solution:

(i) Whole numbers less than 6

Number less than 6 means number to the left side of 6.



Dark shaded dots represent the required whole numbers.
So, 0, 1, 2, 3, 4, 5 are the whole numbers less than 6.

(ii) Whole numbers greater than 4

Dark shaded dots represent the required whole numbers.
So, 5, 6, 7, ... are the required whole numbers.

Represent whole numbers $\geq$ (or $\leq$) a given whole number on a number line

Let us represent whole number $\geq$ (or $\leq$) a given whole number on a number line with the help of an example.

Example. Represent the following whole number on a number line.

(i) Whole numbers less than or equal to 7
(ii) Whole numbers greater than or equal to 3

Solution: (i) whole number ≤ 7

Dark shaded points represent the required whole numbers.

So, the whole numbers less than or equal to 7 are 0, 1, 2, 3, 4, 5, 6, 7.

(ii) Whole number ≥ 3

Dark shaded points represent the required whole numbers with continuation.

So, the whole numbers greater than or equal to 3 are 3, 4, 5, 6, 7, ...

Teacher's Note *Teacher should give more questions to students on blackboard to represent whole numbers on number line.*

Represent whole numbers > but < a given whole number on a number line

Let us represent whole numbers > but < a given whole number on a number line with the help of an example.

Example. Represent the even whole numbers greater than 1 but less than 13 on number line.

Solution: Even whole numbers > 1 but < 13

Dark shaded points represent the required even whole numbers.

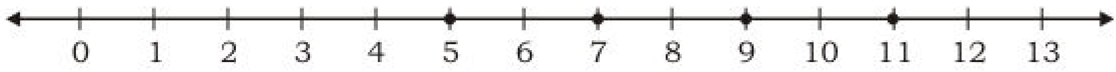
Hence the required even numbers are 2, 4, 6, 8, 10, 12.

Represent whole numbers $\geq$ but $\leq$ a given whole number on a number line

Let us represent whole numbers $\geq$ but $\leq$ a given whole number on a number line with the help of an example.

Example: Represent odd whole numbers greater than or equal to 5 but less than or equal to 11 on a number line.

Solution: Odd whole numbers ≥ 5 but ≤ 11



Dark shaded points represent the required odd whole numbers.

Hence the required odd whole numbers are 5, 7, 9, 11.

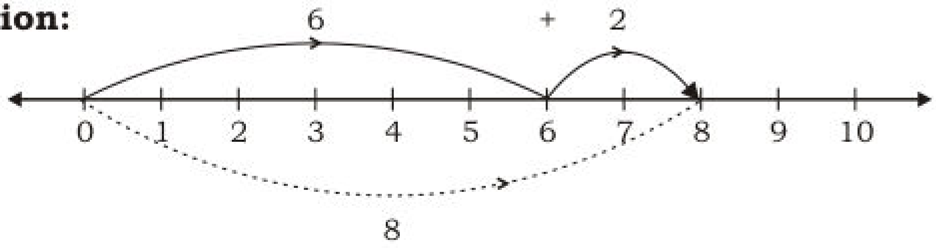
Represent sum of two or more given whole numbers on the number line

The method of adding two or more given whole numbers on the number line is explained with the help of following example.

Teacher's Note *Teacher should draw the number line on blackboard and move a student in front of it to show the sum of numbers on number line.*

Example 1. Find the sum of 6 and 2 by using number line.

Solution:



Its procedure is described below:

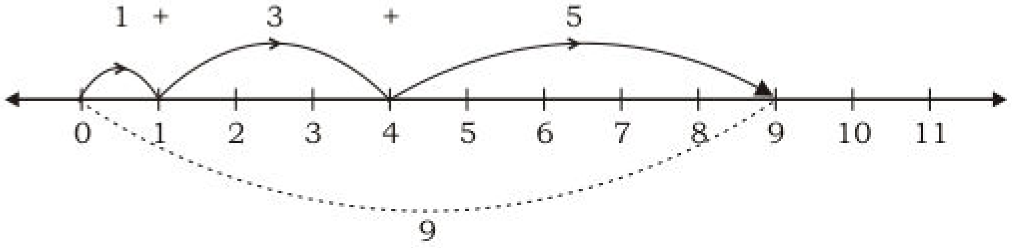
Start from 0, and move 6 units on the right. Again, starting from 6 move 2 units on the right. We reach at number 8.

Hence, 6 + 2 = 8

Example 2.

With the help of a number line, find the sum of 1 + 3 + 5

Solution:



Therefore, 1 + 3 + 5 = 9

EXERCISE 2.1

1. **Write, if possible:**
   (i) The smallest natural number
   (ii) The smallest whole number
   (iii) The largest natural number
   (iv) The largest whole number
2. **Write first ten natural numbers.**
3. **Write first ten whole numbers.**

4. **Represent on number line:**
   - (i) Whole numbers > 8
   - (ii) Whole numbers < 8
   - (iii) Whole numbers ≥ 8
   - (iv) Whole numbers ≤ 8
   - (v) Whole numbers > 3 but < 15
   - (vi) Whole numbers > 5 but ≤ 12
   - (vii) Whole numbers ≥ 4 but ≤ 11
   - (viii) Whole numbers ≥ 3 but < 9
   - (ix) Odd whole numbers greater than 3
   - (x) Even whole numbers greater than or equal to 8 but less than 16.

5. **Find sum of the following whole numbers by using number line.**
   - (i) 1 and 5
   - (ii) 6 and 3
   - (iii) 10 and 2
   - (iv) 8 and 4
   - (v) 2, 3 and 5
   - (vi) 3, 2 and 4

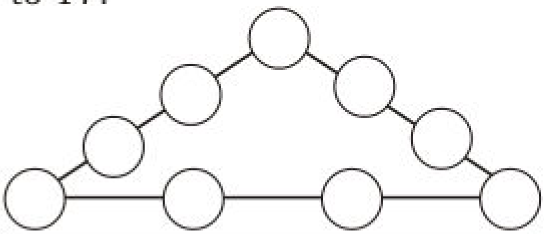
2.2 ADDITION AND SUBTRACTION OF WHOLE NUMBERS

Add and subtract two given whole numbers

We already know how to add and subtract two natural numbers.

Same methods are applied for whole numbers.

How to fill the triangle with 1, 2, 3, 4, 5, 6, 7, 8 and 9 to get the sum of each side equal to 17.



For example, the sum of 5 and 14 is 19 which is again a whole number. Therefore we can say that the sum of whole numbers is always a whole number.

Similarly, 25941 + 58723 = 84664 which is a whole number.

> The sum of two whole numbers is always a whole number.

When we subtract 23 from 53, we get 30. Here 23, 53 and 30 are all whole numbers. Hence, the resulting number in subtraction of whole numbers may or may not be a whole number.

Example 1: Add: 389 and 245

$$\begin{array}{r} {}^{1}3\;{}^{1}8\;9 \\ +\;\;2\;\;4\;\;5 \\ \hline \mathbf{6\;\;3\;\;4} \\ \hline \end{array}$$

As,
$9 + 5 = 14$
$1 + 8 + 4 = 13$
$1 + 3 + 2 = 6$

Hence 389 + 245 = **634**

Example 2: Subtract 535 from 1000

$$\begin{array}{r} {}^{0}1\;{}^{9}\cancel{0}\;{}^{9}\cancel{0}\;{}^{1}0 \\ -\;\;\;5\;\;3\;\;5 \\ \hline \mathbf{0\;\;4\;\;6\;\;5} \\ \hline \end{array}$$

As,
$10 - 5 = 5$
$9 - 3 = 6$
$9 - 5 = 4$
$0 - 0 = 0$

Hence 1000 – 535 = **465**

Example 3: Aslam earns Rs 56835 in a month and his wife earns Rs 35600. Their monthly expenditure is Rs 65000. Find their total monthly income and saving.

Solution: Adding the monthly income of Aslam and his wife.

Monthly income of Aslam	=	56835 rupees
Monthly income of his wife	=	+ 35600 rupees
Total of monthly income	=	92435 rupees

Now, in order to find monthly saving, we subtract monthly expenditure from their total income.

Monthly income	=	92435 rupees
Monthly expenditure	=	65000 rupees
Monthly saving	=	27435 rupees

Hence, their monthly income is Rs 92435 and monthly saving is Rs 27435.

Example 4: Find the sum of smallest five digit number and largest four digit number.

Solution:

Smallest five digit number 10000

Largest four digit number + 9999

19999

Hence, required sum = **19999**

Verify commutative and associative law (under addition) of whole numbers.

(a) Commutative law under addition.

When we add 9 to 30, we get 39 and when we add 30 to 9, then again we get 39. Similarly by adding 100 to 40 and by adding 40 to 100, we get the same whole number 140. In other words, we have

$9 + 30 = 30 + 9$, and $100 + 40 = 40 + 100$

It shows that, for any order of two given whole numbers, their addition give us the same sum. This is known as the *commutative law of whole numbers under addition.*

Thus commutative law under addition is:

> The sum of two whole numbers, in any order, is always same.

Example:

Verify commutative law under addition for whole numbers 85 and 95.

Solution: By Commutative Law under addition, $85 + 95 = 95 + 85$

LHS $= 85 + 95$
$= 180$

85
+ 95
180

RHS $= 95 + 85$
$= 180$

95
+ 85
180

Since, LHS = RHS, therefore $85 + 95 = 95 + 85$

Hence, Commutative law under addition is verified.

(b) Associative law under addition.

We can add only two whole numbers at a time. So, we can get the sum of three whole numbers, say 3, 7 and 12 by two ways:

$\boxed{3+7}+12$	or	$3+\boxed{7+12}$
$= 10 + 12$		$= 3 + 19$
$= 22$		$= 22$

Therefore we can say
$(3 + 7) + 12 = 3 + (7 + 12)$

Hence, three whole numbers added in any order give same result.
This is known as associative law under addition.
Thus Associative law under addition is:

The sum of three whole numbers, in any order, is always same.

Example:
Verify Associative law under addition for 23, 59 and 87.

Solution:
By Associative Law under addition, $(23 + 59) + 87 = 23 + (59 + 87)$

$$\begin{aligned} \text{LHS} &= (23 + 59) + 87 \\ &= 82 + 87 \\ &= 169 \end{aligned} \qquad \begin{array}{r} 23 \\ +\ 59 \\ \hline 82 \\ +\ 87 \\ \hline \mathbf{169} \\ \hline \end{array}$$

$$\begin{aligned} \text{RHS} &= 23 + (59 + 87) \\ &= 23 + 146 \\ &= 169 \end{aligned} \qquad \begin{array}{r} 59 \\ +\ 87 \\ \hline 146 \\ +\ 23 \\ \hline \mathbf{169} \\ \hline \end{array}$$

Since, LHS = RHS, therefore $(23 + 59) + 87 = 23 + (59 + 87)$
So, Associative law under addition is verified.

Teacher's Note *Teacher should help the students to understand the concepts of laws of whole numbers under addition.*

Activity 1 Fill in the following blanks by using Commutative and Associative laws of addition.

(i) 56123 + 71045 = 71045 + **56123**

(ii) 24125 + (41625 + 7123) = (24125 + ________) + 7123

(iii) 47813 + __________ = 51623 + __________

(iv) 567 + (__________ + 1784) = (__________ + 962) + __________

Recognize '0' as additive identity

There is a whole number, which has a unique property that no other whole numbers has. The number is 0, and the property is:

When 0 is added to any whole number, the sum is the whole number itself.

For Example: 1 + 0 = 0 + 1 = 1
25 + 0 = 0 + 25 = 25

And the number "0" is called additive identity.

EXERCISE 2.2

1. What is the sum of the largest number of six digits and the smallest number of seven digits?

2. Population of a village is 2700. If 1070 are men and 915 are women, find the number of children.

3. Arif deposited Rs 45800 in his bank account. After a month he withdrew Rs 3500 from it. How much money was left in his account?

4. Kashif had Rs 82000. He gave Rs 6500 to his wife, Rs 10550 to son and Rs 15335 to daughter. How much money was left with him?

5. Verify commutative law under addition for the following.

(i) 7628 and 39780 (ii) 924981 and 228
(iii) 29000 and 10699 (iv) 50102 and 9019854

6. Verify associative law under addition for the following.

(i) 34006, 2389 and 44380 (ii) 583031, 127 and 3405
(iii) 231, 6090 and 25996 (iv) 412, 3007 and 102341

7. Fill in the blanks to make the following statements true.

(i) 5020 + 849 = ________________ + 5020

(ii) 97864 + 0 = ________________

(iii) 749 + 0 = 0 + ______ = ______

(iv) 3971 + (430 + 300) = (3971 + ____________) + 300

(v) 4853 + 93 = 93 + 4853 = ________________

8. Think a number. Double the number and add 9 to it then add the number you started with. Then divide the amount by 3 and also subtract 3 from the quotient. What the number do you get?

2.3 MULTIPLICATION AND DIVISION OF WHOLE NUMBERS

We know that multiplication is repeated addition and division is repeated subtraction.

Multiply and divide two given whole numbers

(a) Multiplication:

The method of multiplying two whole numbers is same as the method of multiplication of natural numbers.

Consider the following multiplications:

$5 \times 4 = 20, \quad 10 \times 153 = 1530, \quad 0 \times 2453 = 0$

In all these examples, the multiplication of two whole numbers gives a whole number.

The product of two whole numbers is always a whole number.

Let us consider the following examples.

Example 1:
Multiply 2483 by 253

Solution:

$$\begin{array}{r} 2483 \\ \times \quad 253 \\ \hline 7449 \\ 124150 \\ 496600 \\ \hline \boxed{628199} \end{array}$$

Hence
$2483 \times 253 = 628199$

Example 2: Zahida purchased 20 bangles at the price of Rs 80 each. Find the total amount paid by Zahida.

Solution:

Number of bangles: 20
Price: × 80

$$\begin{array}{r} 20 \\ \times \ 80 \\ \hline 00 \\ 160\ \times \\ \hline 1600 \end{array}$$

Hence the total amount is Rs 1600.

(b) Division:

If we divide 9 by 2, then the quotient is 4 and the remainder is 1.

$$\begin{array}{r} 4 \\ 2\overline{)\,9} \\ -8 \\ \hline 1 \end{array}$$

Let us consider division of some other whole numbers along with their quotients and remainders.

Dividend	Divisor	Quotient	Remainder
9	3	3	0
7	5	1	2
14	8	1	6
56	7	8	0

Note:
- The quotient of a whole number divided by 1 is the number itself.
- The quotient of zero divided by a non-zero whole number is zero.

Example 1: Divide 435 by 3. Write quotient and remainder.

Solution:

$$\begin{array}{rrl}
 & 145 & \longrightarrow \text{quotient} \\
\text{divisor} \longleftarrow 3 & \overline{)\,435} & \longrightarrow \text{dividend} \\
 & -3 & \\ \hline
 & 13 & \\
 & -12 & \\ \hline
 & 15 & \\
 & -15 & \\ \hline
 & 0 & \longrightarrow \text{remainder}
\end{array}$$

Hence, $435 \div 3 = 145$.

Example 2: The cost of 15 books is Rs 600. Find the cost of one book.

Solution: In order to find the cost of one book we divide the total cost Rs 600 by number of books, i.e. 15.

$$\begin{array}{rr}
 & 40 \\
15 & \overline{)\,600} \\
 & -600 \\ \hline
 & \times
\end{array}$$

So, the cost of one book is Rs 40.

Example 3:

Find the greatest 4 digit number which is exactly divisible by 23.

Solution:

As greatest 4 digit number is 9999

$$\begin{array}{rrl}
 & 434 & \\
23 & \overline{)\,9999} & \\
 & -92 & \\ \hline
 & 79 & \\
 & -69 & \\ \hline
 & 109 & \\
 & -92 & \\ \hline
 & 17 & = \text{remainder}
\end{array}$$

Thus the required number $= 9999 - 17$

$= 9982$

EXERCISE 2.3

1. **Find the following products of whole numbers.**
   (i) 854×96 (ii) 736×103
   (iii) 256×1008 (iv) 995×158
2. **Divide and find the quotient and remainder.**
   (i) $7772 \div 58$ (ii) $96324 \div 245$
   (iii) $16025 \div 1000$ (iv) $92845 \div 300$
3. A gardener plans to plant 570 trees in 19 rows. Each row should contain equal number of trees. How many trees will be in each row?
4. The shopkeeper purchased 125 television sets. If the cost of each set is Rs 9820, find the cost of all sets.
5. Find the greatest 5 digit number which is exactly divisible by 75.
6. Multiply the greatest number of four digits with the smallest number of three digits.

Verify commutative and associative law (under multiplication) of whole numbers

(a) Commutative law under multiplication

Consider the product of two whole numbers 5 and 6, which is $5 \times 6 = 30$. Also $6 \times 5 = 30$.

Thus 5 multiplied by 6 is the same as 6 multiplied by 5.

Similarly, $401 \times 98 = 98 \times 401$ gives $39298 = 39298$, $3756 \times 23 = 23 \times 3756$ gives $86388 = 86388$, $101 \times 0 = 0 \times 101$ gives $0 = 0$ and $255 \times 1 = 1 \times 255$ gives $255 = 255$.

Hence, two whole numbers multiplied in any order gives equal product. This is known as commutative law under multiplication.

> The product of two whole numbers, in any order, is always same.

Associative law under multiplication

Consider the product of three whole numbers 3, 2 and 4. If we first multiply 3 and 2 and multiply the resulting number by 4, we get

$$(3 \times 2) \times 4 = 6 \times 4 = 24.$$

On the other hand, if we multiply 2 with 4 and multiply their product with 3.

we get $3 \times (2 \times 4) = 3 \times 8 = 24$

Thus, $(3 \times 2) \times 4 = 3 \times (2 \times 4)$

> What are the things we can do without any definite order in our daily life?

Hence, three whole numbers multiplied in any order gives equal product. This is the associative law under multiplication.

Thus associative law under multiplication is:

> The product of three whole numbers, in any order, is always same.

Recognize '1' as multiplicative identity

The whole number 1 has a special role in multiplication just as the number 0 has in the case of addition. As $3 \times 1 = 1 \times 3 = 3$ and $1 \times 5 = 5 \times 1 = 5$ etc. Therefore 1 is the only whole number with the property that when 1 is multiplied with any whole number in any order, the result is the same whole number.

> So "1" is called multiplicative identity.

Note: The number 0 also has a special property in multiplication. We know that $8 \times 0 = 0$, $0 \times 200 = 0$ and $283450 \times 0 = 0$.

> Therefore when 0 is multiplied with any whole number the result is 0.

2.4 MULTIPLICATION AND ADDITION (SUBTRACTION) OF WHOLE NUMBERS

Verify distributive law of multiplication over addition

For the three whole numbers 2, 4 and 7. First we consider $4 + 7$ and then $2 \times (4 + 7)$ to get $2 \times 11 = 22$.

Also $(2 \times 4) + (2 \times 7) = 8 + 14 = 22$

Therefore $2 \times (4 + 7) = (2 \times 4) + (2 \times 7)$

> What are the things in our daily life must be done in a certain order?

This is called distributive law of multiplication over addition.

Let us demonstrate the distributive law of multiplication over addition with the help of example $5\,(10 + 8) = 5 \times 10 + 5 \times 8$

Solution:

L.H.S $= 5\,(10 + 8)$	R.H.S $= 5 \times 10 + 5 \times 8$
$= 5\,(18)$	$= 50 + 40$
$= 90$	$= 90$

$$\because \text{L.H.S} = \text{R.H.S}$$

$$\therefore 5\,(10 + 8) = 5 \times 10 + 5 \times 8$$

Hence distributive law of multiplication over addition is verified.

Verify distributive law of multiplication over subtraction (with positive difference)

For the above three whole numbers 2, 4 and 7,

$$2 \times (7 - 4) = 2 \times 3 = 6 \text{ and } (2 \times 7) - (2 \times 4) = 14 - 8 = 6$$

Therefore, $2 \times (7 - 4) = (2 \times 7) - (2 \times 4)$

This is called distributive law of multiplication over subtraction. Let us demonstrate of Distributive law of multiplication over subtraction with help of example $5\,(20 - 2) = 5 \times 20 - 5 \times 2$

Solution:

L.H.S	= 5 (20 – 2)	R.H.S	= 5 × 20 – 5 × 2
	= 5 (18)		= 100 – 10
	= 90		= 90

$$\because \text{L.H.S} = \text{R.H.S}$$

$$\therefore 5\,(20 - 2) = 5 \times 20 - 5 \times 2$$

Hence distributive law of multiplication under subtraction is verified.

EXERCISE 2.4

1. Fill in the blanks.

(i) $5 \times 0 =$ ________

(ii) $6 \times 1 =$ ________

(iii) $2 \times 5 = 5 \times$ ______

(iv) $3 \times (1 \times 4) = (3 \times 1) \times$ _____

(v) $5 \times ($ _____ $\times 8) = ($ _____ $\times 7) \times 8$

(vi) $541 \times (645 +$ ______ $) = 541 \times 645 + 541 \times 964$

(vii) $345 \times (650 - 125) = 345 \times 650$ ______ $345 \times$ ______

2. Write True or False

(i) $545 \times 248 = 545 + 248$ ________

(ii) $67 - (125 - 12) = (67 - 125) - 12$ ________

(iii) $64 \times (245 - 10) = 64 \times 245 - 64 \times 10$ ________

(iv) $95 \times (25 \times 14) = (95 \times 25) + 14$ ________

3. **Name and verify the law used in each of the following.**

(i) $9 \times 7 = 7 \times 9$ (ii) $4 \times (6 \times 3) = (4 \times 6) \times 3$

(iii) $5 \times (7 - 1) = 5 \times 7 - 5 \times 1$ (iv) $2 \times (8 + 3) = 2 \times 8 + 2 \times 3$

4. Verify commutative law under addition for 530 and 750.

5. Verify Associative Law under multiplication for 240, 425 and 35.

6. Verify distributive law of multiplication over addition for 300, 615 and 975.

REVIEW EXERCISE 2

1. **Fill in the blanks by using laws under multiplication of whole numbers.**

(i) $2 \times$ ______ $= 3 \times$ ______

(ii) $112 \times 528 =$ ______ $\times$ ______

(iii) $5 \times (3 \times 8) = (5 \times 3) \times$ ______

(iv) ______ $\times\ 7 = 7 \times$ ______ $= 0$

(v) $9 \times (2 + 16) =$ ______ $\times 2 +$ ______ $\times 16$

(vi) $(20 + 4) \times 1 =$ ______ $\times 1 + 4 \times$ ______

(vii) $17 \times (8 - 3) = 17 \times$ ______ $-$ ______ $\times 3$

(viii) $(12 - 6) \times 32 = 12 \times$ ______ $- 6 \times$ ______

2. **Fill in the blanks.**

(i) $5 \times 4 = \square \times 5$

(ii) $15 \times (10 \times 6) = (\square \times 10)$

(iii) $10 \times (15 + 6) = (10 \times 15) + (\square \times 6)$

(iv) $10 \times (100 + 60) = (\square \times 100) + (10 \times \square)$

3. Write True or False.

(i) $5 - 3 = 3 - 5$ (ii) $3 \times 1 + 7 = 3 \times 8$

(iii) $2 - (0 - 8) = (2 - 0) + 8$ (iv) $9 + (7 + 5) = (9 + 7) + 5$

(v) $4 \times (35 \times 2) = (4 \times 35) \times 2$

(vi) $24 - (50 - 6) = (24 - 50) - 6$

(vii) $14 \div 0 = 14$ (viii) $0 \div 125 = 0$

(ix) $18 \div 18 = 0$ (x) $75 \div 75 = 1$

4. Write the predessor and successor of the following numbers:

(i) 671 (ii) 245

5. Choose the correct answer:

(i) The smallest natural number is ______.
(a) 0 (b) 1 (c) 2 (d) 100

(ii) The predecessor of 1 in the set of whole numbers is ______.
(a) 0 (b) 2 (c) 3 (d) none

(iii) The smallest seven digit number is ______.
(a) 1234567 (b) 9999999
(c) 1111111 (d) 1000000

(iv) The greatest six digit number is _________
(a) 876543 (b) 999999 (c) 111111 (d) 100000

(v) The numbers divisible by 2 are called______numbers.
(a) prime (b) even (c) odd (d) whole

6. Draw a number line to represent the following whole numbers.

(i) 0, 1, 3, 9 (ii) Whole numbers > 3

(iii) Whole numbers ≤ 8

(iv) Whole numbers > 5 but < 10

(v) Whole numbers ≥ 1 but ≤ 8

7. Find the sum of 4, 2, 3 and 5 on number line.
8. Find two whole numbers whose sum is:
   (i) 9 (ii) 24 (iii) 31
9. Verify commutative laws of addition and multiplication for 190 and 330.
10. Verify Associative laws of addition and multiplication for 20, 30 and 60.
11. Verify distributive laws of multiplication over addition and subtraction for 700, 500 and 100.
12. What is the cost of 640 photocopies at Rs 1.50 per copy?
13. Find the greatest number of 4 digits which is exactly divisible by 44.

SUMMARY

- Natural numbers are used for counting.
- Some of the set of numbers and their notations are:
  - Set of natural numbers, N = {1, 2, 3, ...}
  - Set of whole numbers, W = {0, 1, 2, ...}
- Sum and product of two whole numbers is also a whole number.
- Addition of whole numbers is commutative and associative.
- Multiplication of whole numbers is commutative and associative.
- Zero is additive identity in the set of whole numbers.
- The multiplicative identity in the set of whole numbers is 1.
- Multiplication is distributive over addition and subtraction (with positive difference) in the set of whole numbers.
- Division of two whole numbers is a whole number only if divisor completely divides the number and remainder is zero.